

AI Hallucination Courtroom Student Handout

TOK Cognitive Science Activity Bank • Knowledge and Technology • 35-45 min

Category	Knowledge and Technology	Time	35-45 min
Difficulty	Extended	ID	ai-hallucination-courtroom

Big idea: Fluency can be mistaken for knowledge.

Student-Facing Prompt

Put the AI paragraph on trial. For each claim, decide: grounded, unsupported, false, or needs verification.

Concrete Stimulus Packet

AI Paragraph On Trial	Students should verify AI-generated answers because large language models do not truly understand facts; they predict likely sequences of words based on patterns in training data. This means they can produce answers that sound fluent and authoritative even when details are wrong. For example, a 2021 study by Elena Marsh and David Kline in the Journal of Digital Reasoning found that 68% of AI-generated school summaries contained at least one factual error. AI tools are also connected to the live internet by default, so they usually update themselves instantly when new information appears. Because of this, students should treat AI as a starting point for inquiry, not as a final authority.	Prompt given to AI: Explain why students should verify AI-generated answers.
Courtroom Roles	Prosecution: argue the paragraph is unreliable. Defence: argue parts are useful if treated carefully. Expert witness: explain hallucinations. Judge: classify claims. Jury: vote mostly reliable, partly reliable, or misleading.	Assign roles before students see the hidden issues.
Hidden Issues	Fabricated citation: Elena Marsh and David Kline, Journal of Digital Reasoning, 2021. Plausible false detail: AI tools are connected to the live internet by default. Grounded claim: AI can produce fluent but incorrect answers. TOK point: fluency can create misplaced trust.	Teacher reveal after verdicts.
Claim Check: AI predicts likely word sequences	Verdict: Grounded	This broadly describes how language models generate text.
Claim Check: AI can sound authoritative while wrong	Verdict: Grounded	This is a known risk of generative AI.
Claim Check: Marsh and Kline study, 2021	Verdict: False/Fabricated	The citation is invented for this activity.

Actual Lesson Questions

#	Question for students
1	Which claim in the paragraph sounds most authoritative?

#	Question for students
2	Which claim can be accepted as broadly grounded?
3	Which claim is fabricated?
4	Which claim is plausible but false or overgeneralised?
5	What made the paragraph feel trustworthy?
6	Can this paragraph count as knowledge? Why or why not?

Ready-to-Use Examples

- An AI-generated paragraph with one fabricated citation and one plausible false detail.
- Courtroom roles for prosecution, defence, expert witness, judge, and jury.
- An evidence table asking which claims are grounded, unsupported, or unverifiable.

Student Task

For each claim, decide: grounded, unsupported, false, or needs verification. Then write: "Can this paragraph count as knowledge? Why or why not?"

Activity Steps

1. Give all groups the same paragraph.
2. Assign roles: prosecution, defence and judge.
3. Prosecution identifies unsupported or false claims. Defence argues how the tool could still be useful. Judges decide what evidence is required.
4. Finish by writing class rules for responsible AI use.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Can fluency be mistaken for knowledge?
- Can a machine be a knower, or only a producer of outputs?
- Who is responsible for AI misinformation?

Knowledge Questions

- Can fluency be mistaken for knowledge?
- Can a machine be a knower, or only a producer of outputs?

- Who is responsible for AI misinformation?

Transfer Example

For example, run this activity with an ai-generated paragraph with one fabricated citation and one plausible false detail. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...